

NAND2TETRIS ASSIGNMENT

Project 1

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Chips Implemented

- 1 Not
- 2 And
- 3 Or
- 4 Xor
- 5 Mux
- 6 DMux
- 7 Not16
- 8 And16
- 9 Or16
- 10 Mux16
- 11 Or8Way
- 12 Mux4Way16
- 13 Mux8Way16
- 14 DMux4Way
- 15 DMux8Way

Chip: Not

NAND2Terns

/ Hardware Simulator memory developer_board list_alt computer code grid_on function menu_book

HDL ☐ --LBSt Project 1 Not **Chip Not** Eval Reset Clock: 0

```
5 /**
7  * if (in) out = 0, else out = 1
8  */
9 CHIP Not {
10     IN in;
11     OUT out;
12
13     PARTS:
14     Nand(a=in, b=in, out=out);
15 }
16
```

Input pins
in 1

Output pins
out 0

Test **Steps: 5 Outputs: 3** Slow Fast

Test Script Compare File Output File Diff Table

```
11 eval,
12 output;
13
14 set in 1,
15 eval,
16 output;
```

Simulation successful: The output file is identical to the compare file

Chip: And

NAND2Tetris

/ Hardware memory developer_board list_alt computer code grid_on function menu_book

Simulator

HDL ☐ --LBSt Project 1 And Eval Reset Clock: 0

```
9 CHIP And {
10   IN a, b;
11   OUT out;
12
13   PARTS:
14     Nand(a=a, b=b, out=nandOut);
15     Not(in=nandOut, out=out);
16 }
17
18
19
```

Input pins

a	1
b	1

Output pins

out	1
-----	---

Internal pins

nandOut	0
---------	---

Test     Steps: 7 Outputs: 5 Slow Fast

Test Script Compare File Output File Diff Table

```
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
```

Simulation successful: The output file is identical to the compare file

Chip: Or

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Simulator

HDL ☐ --LBSt Project 1 Or Eval Reset Clock: 0

```
5 /**
8 */
9 CHIP Or {
10     IN a, b;
11     OUT out;
12
13     PARTS:
14         Nand(a=a, b=a, out=nota);
15         Nand(a=b, b=b, out=notb);
16         Nand(a=nota, b=notb, out=out);
17 }
```

Input pins

a 1

b 1

Output pins

out 1

Internal pins

nota 0

Test     Steps: 7 Outputs: 5 Slow Fast

Test Script Compare File Output File Diff Table

```
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
```

Simulation successful: The output file is identical to the compare file

Chip: Xor

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HDL ☐ --LBSt Project 1 Xor

```
5 /**
9 CHIP Xor {
10     IN a, b;
11     OUT out;
12
13     PARTS:
14         Nand(a=a, b=b, out=nand1);
15         Nand(a=a, b=nand1, out=nand2);
16         Nand(a=b, b=nand1, out=nand3);
17         Nand(a=nand2, b=nand3, out=out)
18 }
```

Test    **Steps: 7 Outputs: 0** Slow  Fast

Test Script Compare File Output File Diff Table

```
3 // by Nisan and Schocken, MIT Press.
4 // File name: projects/1/Xor.tst
5
6 Load Xor.hdl,
7 compare-to Xor.cmp,
```

Chip: Mux

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Simulator

HDL --LBSt Project 1 Mux

Chip Mux Eval Reset Clock: 0

```
5 /**
9 CHIP Mux {
10     IN a, b, sel;
11     OUT out;
12
13     PARTS:
14         Not(in=sel, out=nsel);
15         And(a=a, b=nsel, out=anda);
16         And(a=b, b=sel, out=andb);
17         Or(a=anda, b=andb, out=out);
18 }
```

Input pins
a 1
b 0
sel 1

Output pins
out 0

Internal pins

Test **Steps: 11 Outputs: 9** Slow Fast

Test Script Compare File Output File Diff Table

```
43 eval,
44 output;
45
46 set sel 1,
47 eval,
48 output;
```

Simulation successful: The output file is identical to the compare file

Chip: DMux

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/ Hardware memory developer_board list_alt computer code grid_on function menu_book

Simulator

HDL ☐ --LBSt Project 1 DMux Chip DMux Eval Reset Clock: 0

```
10 CHIP DMux {
11     OUT a, b;
12     PARTS:
13     Not(in=sel, out=nSel);
14     And(a=in, b=nSel, out=a);
15     And(a=in, b=sel, out=b);
16 }
```

Input pins

in 1

sel 1

Output pins

a 0

b 1

Internal pins

Test   Steps: 7 Outputs: 5 Slow Fast

Test Script  Compare File Output File Diff Table

```
21 eval,
22 output;
23
24 set sel 1,
25 eval,
26 output;
```

Simulation successful: The output file is identical to the compare file

Chip: Not16

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HDL ☐ --LBSt Project 1 Not16 **Chip Not16** Eval Reset Clock: 0

```
10 CHIP Not16 {
14     PARTS:
15     Not(in=in[0], out=out[0]);
16     Not(in=in[1], out=out[1]);
17     Not(in=in[2], out=out[2]);
18     Not(in=in[3], out=out[3]);
19     Not(in=in[4], out=out[4]);
20     Not(in=in[5], out=out[5]);
21     Not(in=in[6], out=out[6]);
22     Not(in=in[7], out=out[7]);
23     Not(in=in[8], out=out[8]);
```

Input pins
in 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec

Output pins
out 1 1 1 0 1 1 0 1 1 1 0 0 1 0 1 1 dec

Test Steps: 8 Outputs: 6 Slow Fast

Test Script Compare File Output File Diff Table

```
23 eval,
24 output;
25
26 set in %00001001000110100,
27 eval,
28 output;
```

Simulation successful: The output file is identical to the compare file

Chip: And16

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HDL ☐ --LBSt Project 1 And16

Chip And16 Eval Reset Clock: 0

```
10 CHIP And16 {
11
12     PARTS:
13
14     And(a=a[0], b=b[0], out=out[0]);
15     And(a=a[1], b=b[1], out=out[1]);
16     And(a=a[2], b=b[2], out=out[2]);
17     And(a=a[3], b=b[3], out=out[3]);
18     And(a=a[4], b=b[4], out=out[4]);
19     And(a=a[5], b=b[5], out=out[5]);
20     And(a=a[6], b=b[6], out=out[6]);
21     And(a=a[7], b=b[7], out=out[7]);
22     And(a=a[8], b=b[8], out=out[8]);
23 }
```

Input pins

a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec

b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec

Output pins

out 0 0 0 1 0 0 0 0 0 0 0 1 1 0 1 0 0 dec

Test Steps: 9 Outputs: 7 Slow Fast

Test Script Compare File Output File Diff Table

```
33 output;
34
35 set a %B0001001000110100,
36 set b %B1001100001110110,
37 eval,
38 output;
```

Successful: The output file is identical to the compare file

<https://nand2tetris.org>

Chip: Or16

NAND2Tetris

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Simulator

HDL ☐ --LBSt Project 1 Or16 **Chip Or16** Eval Reset Clock: 0

```
10 CHIP Or16 {
14     PARTS:
15     Or(a=a[0], b=b[0], out=out[0]);
16     Or(a=a[1], b=b[1], out=out[1]);
17     Or(a=a[2], b=b[2], out=out[2]);
18     Or(a=a[3], b=b[3], out=out[3]);
19     Or(a=a[4], b=b[4], out=out[4]);
20     Or(a=a[5], b=b[5], out=out[5]);
21     Or(a=a[6], b=b[6], out=out[6]);
22     Or(a=a[7], b=b[7], out=out[7]);
23     Or(a=a[8], b=b[8], out=out[8]);
```

Input pins

a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec

b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec

Output pins

out 1 0 0 1 1 0 1 0 0 1 1 1 0 1 1 0 dec

Test Steps: 9 Outputs: 7 Slow Fast

Test Script Compare File Output File Diff Table

```
33 output;
34
35 set a %0001001000110100,
36 set b %01001100001110110,
37 eval,
38 output;
```

Simulation successful: The output file is identical to the compare file

Chip: Mux16

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Simulator

HDL ☐ --LBSt Project 1 Mux16 **Chip Mux16** Eval Reset Clock: 0

```
10 CHIP Mux16 {
14   PARTS:
15     Mux(a=a[0], b=b[0], sel=sel, out=out[0]);
16     Mux(a=a[1], b=b[1], sel=sel, out=out[1]);
17     Mux(a=a[2], b=b[2], sel=sel, out=out[2]);
18     Mux(a=a[3], b=b[3], sel=sel, out=out[3]);
19     Mux(a=a[4], b=b[4], sel=sel, out=out[4]);
20     Mux(a=a[5], b=b[5], sel=sel, out=out[5]);
21     Mux(a=a[6], b=b[6], sel=sel, out=out[6]);
22     Mux(a=a[7], b=b[7], sel=sel, out=out[7]);
23     Mux(a=a[8], b=b[8], sel=sel, out=out[8]);
```

Input pins
a 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec
b 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
sel 1
Output pins
out 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec

Test Steps: 11 Outputs: 9 Slow Fast

Test Script Compare File Output File Diff Table

```
43 eval,
44 output;
45
46 set sel 1,
47 eval,
48 output;
```

Simulation successful: The output file is identical to the compare file

Chip: Or8Way

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Simulator

HDL ☐ --LBSt Project 1 Or8Way **Chip Or8Way** Eval Reset Clock: 0

```
9 CHIP Or8Way {
12
13     PARTS:
14         Or(a=in[0], b=in[1], out=o1);
15         Or(a=in[2], b=in[3], out=o2);
16         Or(a=in[4], b=in[5], out=o3);
17         Or(a=in[6], b=in[7], out=o4);
18         Or(a=o1, b=o2, out=o5);
19         Or(a=o3, b=o4, out=o6);
20         Or(a=o5, b=o6, out=out);
21 }
```

Input pins
in 0 0 1 0 0 1 1 0 dec

Output pins
out 1

Internal pins
o1 1
o2 1

Test Steps: 8 Outputs: 6 Slow Fast

Test Script Compare File Output File Diff Table

```
23 eval,
24 output;
25
26 set in %B00100110,
27 eval,
28 output;
```

Simulation successful: The output file is identical to the compare file

Chip: Mux4Way16

IAND2Tetris
Hardware memory developer_board list_alt computer code grid_on function menu_book

HDL --LBSt Project 1 Mux4Way16

Chip Mux4Way16 Eval Reset Clock: 0

```
5 /**
11 */
12 CHIP Mux4Way16 {
13     IN a[16], b[16], c[16], d[16], sel[2];
14     OUT out[16];
15
16     PARTS:
17         Mux16(a=a, b=b, sel=sel[0], out=ab);
18         Mux16(a=c, b=d, sel=sel[0], out=cd);
19         Mux16(a=ab, b=cd, sel=sel[1], out=out);
20 }
```

Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec
b 1 0 0 1 1 0 0 0 0 0 1 1 1 0 1 0 dec
c 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec
d 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 dec
sel 1 1 dec

Test Steps: 11 Outputs: 9 Slow Fast

Test Script Run Compare File Output File Diff Table

```
43 eval,
44 output;
45
46 set sel 3,
47 eval,
48 output;
```

Simulation successful: The output file is identical to the compare file

Chip: Mux8Way16

IAND2Tetris
Hardware memory developer_board list_alt computer code grid_on function menu_book
simulator

HDL --LBSt Project 1 Mux8Way16

Chip Mux8Way16 Eval Reset Clock: 0

```
16 CHIP Mux8Way16 {
21
22     PARTS:
23     Mux4Way16(a=a, b=b, c=c, d=d,
24               sel=sel[0..1], out=abcd);
25
26     Mux4Way16(a=e, b=f, c=g, d=h,
27               sel=sel[0..1], out=efgh);
28
29     Mux16(a=abcd, b=efgh, sel=sel[2], out=out);
30 }
```

Input pins
a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec
b 0 0 1 0 0 0 1 1 0 1 0 0 0 1 0 1 dec
c 0 0 1 1 0 1 0 0 0 1 0 1 0 1 1 0 dec
d 0 1 0 0 0 1 0 1 0 1 1 0 0 1 1 1 dec
e 0 1 0 1 0 1 1 0 0 1 1 1 1 0 0 0 dec

Test **Steps: 19 Outputs: 17** Slow Fast

Test Script Compare File Output File Diff Table
83 eval,
84 output;
85
86 set sel 7,
87 eval,
88 output;

Simulation successful: The output file is identical to the compare file

Chip: DMux4Way

The screenshot displays the Nand2Tetris simulator interface. At the top, a navigation bar includes links for Hardware, memory, developer_board, list_alt, computer, code, grid_on, function, and menu_book. Below this, a toolbar contains icons for HDL, Project 1, DMux4Way, and a folder icon. The main area is divided into three sections:

- Code Editor:** Displays Verilog code for a 4-to-1 multiplexer (DMux4Way). The code includes a module definition with inputs 'in' and 'sel[2]', outputs 'a', 'b', 'c', and 'd', and a 'PARTS' section for internal components.
- Chip DMux4Way:** A control panel for the multiplexer. It features 'Eval', 'Reset', and 'Clock: 0' buttons. It shows 'Input pins' with 'in' set to 1 and 'sel' set to 11 (with a 'dec' button). It also shows 'Output pins' with 'a', 'b', and 'c' all set to 0.
- Test Console:** A section for running tests. It includes a 'Test' button, a 'Test Script' button, and a 'Run' button. Below these are fields for 'Compare File', 'Output File', and 'Diff Table'. The console shows the execution of a test script with commands like 'eval', 'output', 'set sel %B11', 'eval', and 'output'.

At the bottom, a green status bar indicates: "Simulation successful: The output file is identical to the compare file".

Chip: DMux8Way

NAND2Tetris
/ Hardware memory developer_board list_alt computer code grid_on function menu_book
Simulator

HDL ☐ --LBSt Project 1 DMux8Way Chip DMux8Way Eval Reset Clock: 0

```
16 CHIP DMux8Way {
19
20   PARTS:
21     DMux(in=in, sel=sel[2], a=abcd, b=efgh);
22
23     DMux4Way(in=abcd, sel=sel[0..1],
24               a=a, b=b, c=c, d=d);
25
26     DMux4Way(in=efgh, sel=sel[0..1],
27               a=e, b=f, c=g, d=h);
28 }
```

Input pins
in 1
sel 1 1 1 dec

Output pins
a 0
b 0
c 0

Test Steps: 19 Outputs: 17 Slow Fast

Test Script Compare File Output File Diff Table

```
67 eval,
70 output;
71
72 set sel %B111,
73 eval,
74 output;
```

Simulation successful: The output file is identical to the compare file

Project 2

Chip: ALU

The screenshot displays the NAND2Tetris web IDE interface. The top navigation bar includes links for memory, developer_board, list_alt, computer, code, grid_on, function, and menu_book. The main workspace is divided into three panels:

- Left Panel (HDL):** Shows the Verilog code for the ALU chip. The code is organized into two parts: `Step 1: Handle zx and nx (zero/negate x)` and `Step 2: Handle zy and ny (zero/negate y)`. It uses `Mux16` and `Not16` components to implement the logic.
- Right Panel (Chip ALU):** Displays the input pins for the ALU chip. The pins are `x`, `y`, `zx`, `nx`, and `zy`. The values are `x=000000000000010001`, `y=000000000000000011`, `zx=0`, `nx=1`, and `zy=0`. Each pin has a `dec` button for decimal input.
- Bottom Panel (Test):** Shows the test script and the simulation results. The test script includes commands like `set zy 0`, `set ny 1`, `set f 0`, `set no 1`, `eval`, and `output`. The simulation results show that the output file is identical to the compare file, indicating a successful simulation.

The simulation status bar at the bottom indicates: "Simulation successful: The output file is identical to the compare file".

Chip: HalfAdder

NAND2Tetris / Hardware memory developer_board list_alt computer code grid_on function menu_book
Simulator

HDL ☐ --LBSt Project 2 HalfAdder **Chip HalfAdder** Eval Reset Clock: 0

```
5 /**
7 */
8 CHIP HalfAdder {
9     IN a, b;    // 1-bit inputs
10    OUT sum,    // Right bit of a + b
11        carry; // Left bit of a + b
12
13    PARTS:
14    Xor(a=a, b=b, out=sum);
15    And(a=a, b=b, out=carry);
16 }
```

Input pins
a 1
b 1
Output pins
sum 0
carry 1

Test Steps: 7 Outputs: 5 Slow Fast

Test Script Co Reset File Output File Diff Table

```
23 output;
24
25 set a 1,
26 set b 1,
27 eval,
28 output;
```

Simulation successful: The output file is identical to the compare file

Chip: FullAdder

NAND2Tetris

/ Hardware memory developer_board list_alt computer code grid_on function menu_book

Simulator

HDL ☐ --LBSt Project 2 FullAdder Chip FullAdder Eval Reset Clock: 0

```
5 /**
8 CHIP FullAdder {
9   IN a, b, c; // 1-bit inputs
10  OUT sum,    // Right bit of a + b + c
11      carry;  // Left bit of a + b + c
12
13  PARTS:
14    HalfAdder(a=a, b=b, sum=sum1, carry=carry1);
15    HalfAdder(a=sum1, b=c, sum=sum, carry=carry2);
16    Or(a=carry1, b=carry2, out=carry);
17 }
```

Input pins

a 1

b 1

c 1

Output pins

sum 1

carry 1

Test Steps: 11 Outputs: 9 Slow Fast

Test Script Compare File Output File Diff Table

```
41 eval,
42 output;
43
44 set c 1,
45 eval,
46 output;
```

Simulation successful: The output file is identical to the compare file

Chip: Add16

NAND2Tetris

/ Hardware memory developer_board list_alt computer code grid_on function menu_book

Simulator

HDL ☐ --LBSt Project 2 Add16

```
9 CHIP Add16 {
22   FullAdder(a=a[7], b=b[7], c=c7, sum=out[7], carry=c8);
23   FullAdder(a=a[8], b=b[8], c=c8, sum=out[8], carry=c9);
24   FullAdder(a=a[9], b=b[9], c=c9, sum=out[9], carry=c10);
25   FullAdder(a=a[10], b=b[10], c=c10, sum=out[10], carry=c11);
26   FullAdder(a=a[11], b=b[11], c=c11, sum=out[11], carry=c12);
27   FullAdder(a=a[12], b=b[12], c=c12, sum=out[12], carry=c13);
28   FullAdder(a=a[13], b=b[13], c=c13, sum=out[13], carry=c14);
29   FullAdder(a=a[14], b=b[14], c=c14, sum=out[14], carry=c15);
30   FullAdder(a=a[15], b=b[15], c=c15, sum=out[15], carry=c16);
31 }
```

Chip Add16 Eval Reset Clock: 0

Input pins

a 0 0 0 1 0 0 1 0 0 0 1 1 0 1 0 0 dec

b 1 0 0 1 1 0 0 0 0 1 1 1 0 1 1 0 dec

Output pins

out 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 dec

Internal pins

cl 0

Test Steps: 9 Outputs: 7 Slow Fast

Test Script Run Compare File Output File Diff Table

```
33 output;
34
35 set a %B0001001000110100,
36 set b %B1001100001110110,
37 eval,
38 output;
```

Simulation successful: The output file is identical to the compare file